PanskuraBanamali College (Autonomous)

M.Sc.1st Semester Examination 2022 (Online mode)

Subject :Computer Science Paper: COS-102

Paper Name : Advanced Computer architecture

F.M. : 40 Time : 2 Hours

1.Answer any two questions: 2 x 04 =8

a) What is parallel processing? Why parallel process is needed?
b) What is overlapped windows register? Calculate windows size where use 10 global and local register,6 are common register required in 4 window.
c) Difference between serial and parallel processing. Difference between RISC and CISC

2. Answer any two questions: 2 x 08 =16

a) What is hazard? State different types of hazard.
The time of a process of a sub operation in each segment tp=20 ns. The pipeline has k=45 executes n=100 task in sequence.
i) What time k to need the pipe line to complete the process?
ii) calculate the speed up ratio.
5+3

b) Discuss 4 segment arithmetic pipelines. with example.
Describe the difference steps of instruction pipeline 5+3

c) What is array processor? State different types of array processor.
Difference between Array and vector processing.
6+2

3. Answer any one question: 1 x 16 =16

a) State VLIW processor Architecture.
Difference between Super scaler and VLIW preprocessor.
What is Locality of reference?
Describe RISC and CISC. 5+3+2+6

b) What is Distributed processing?
State Flynn's classification of computer architecture
Difference between Isolated IO and Memory mapped IO
Write a note on microprocessor instructions set. 3+4+3+6

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